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REVIEW ARTICLE

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Photobiomodulation for diabetes and its complications: a review of general presentation, mechanisms and efficacy

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ABSTRACT

Diabetes mellitus is a metabolic disease that is marked by persistent hyperglycemia due to inadequate insulin secretion or insulin resistance. Its prevalence is increasing yearly. Diabetes mellitus can lead to serious health complications that are the primary cause of mortality and disability among diabetic patients, including diabetic retinopathy, diabetic foot ulcers, diabetic peripheral neuropathy, and diabetic periodontitis, and so on. Traditional treatments for diabetes and its complications still suffer from limited clinical efficacy and high therapeutic side effects. Photobiomodulation (PBM), which utilizes low levels of red or near-infrared laser to irradiate cells and tissues, has been shown to be efficacious for a wide range of organ damage. In this study, we focus on the application of PBM in diabetes and its complications and mechanisms, as well as the advantages, disadvantages with the aim of developing new ideas for the application of PBM.

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1. Introduction

Diabetes mellitus, a disease of the endocrine system caused by insulin deficiency or insulin resistance and characterized by abnormal blood glucose levels, has become one of the most serious and common chronic diseases of our time. It is projected that by 2050, approximately 1.31 billion individuals will be affected by diabetes [1]. The World Health Organization anticipates that diabetes will rank as the seventh leading cause of death worldwide by 2030 [2]. The numerous complications of diabetes are the primary causes of disability and even death, such as macrovascular disease and microvascular disease. Most diabetic patients require lifelong use of glucose-lowering medications or insulin injections, which are effective in treating the hyperglycemia of diabetes but do not completely prevent the onset and progression of its complications. Additionally, they can be expensive and difficult for patients to consistently persist. Therefore, it is particularly important to provide another practical option for people with diabetes.

Some researchers have defined PBM, also known as low-level laser therapy (LLLT), as a form of light therapy utilizing non-ionizing light sources, such as lasers, light-emitting diodes (LEDs), and broad-spectrum light, which operate within the visible and infrared spectra [3,4]. The wavelengths of PBM typically range from 600 nm to 700 nm and 780 nm to 1100 nm, with power densities typically ranging from 5 mW cm^{-2} to 5 W cm^{-2}. These levels are lower than those employed for tissue ablation, cutting, and thermal coagulation, hence, termed 'low level' [5]. It applies light to the skin, acupuncture points, etc., stimulating biochemical changes in cells, aiding tissue repair, and reducing inflammation. Currently, PBM is most commonly utilized in clinical settings for wound regeneration, non-healing ulcers, and pain management [6]. This review mainly focuses on the current progress in applying PBM for diabetes and its complications. Specifically, we have delved into various aspects of PBM, including dosage, wavelength, mechanisms, and experiment or clinical outcomes. Our review integrates

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findings from cell-based research, animal models, and human trials, encompassing both positive and negative results, while excluding studies utilizing methods other than PBM. Through this review, we aim to illuminate the current progress and future directions of PBM therapy in diabetes management.

2. Mechanism of action of PBM in high glucose environment

Current research suggests that PBM produces a chemical effect on tissue cells rather than a thermal effect [7]. The mechanisms of PBM are summarized in Figure 1. It acts mainly through cytochrome c oxidase (CCO) and light or heat activated transient receptor (TRPV) [8]. In the case of 600-810 nm red laser, the chromophore that absorbs the photons is the CCO [9], which leads to upregulation of cellular respiration, a significant increase in reactive oxygen species (ROS), nitric oxide (NO), adenosine triphosphate (ATP) levels and cyclic adenosine monophosphate (cAMP) [10–12]. At the wavelengths of 800–1064 nm, the photosensitive channel TRPV absorbs most of the photons, resulting in the opening of Ca^{2+} channel on the cell membrane and an increment of intracellular Ca^{2+} [13].

NO is another important molecule for PBM function. PBM treatment has been shown to raise serum NO concentration, reduce lipid peroxidation in aortic rings, and prevent elevation on blood pressure in animals fed with high-fat diet (HFD) [14]. NO and O_2 vie for occupancy of the identical binding site in CCO [15]. Infrared ray (IR) and Near infrared ray (NIR) photons from low-level lasers are absorbed by NO, leading to NO photolysis or detachment from CCO, releasing oxygen binding sites and promoting ATP production [16]. In recent years, a new theory for NO production has been proposed, where CCO is able to act as a nitrite reductase under low oxygen conditions to reduce nitrite to NO [17]. The NO produced is a potent vasodilator, causing vasodilation to allow more immune cells to enter the tissue. And it also involves enhancing endothelial function through its antioxidant effects, improving NO bioavailability and/or signaling pathway [14]. PBM also presented the analgesic effect through activation of the L-arginine/NO pathway [18].

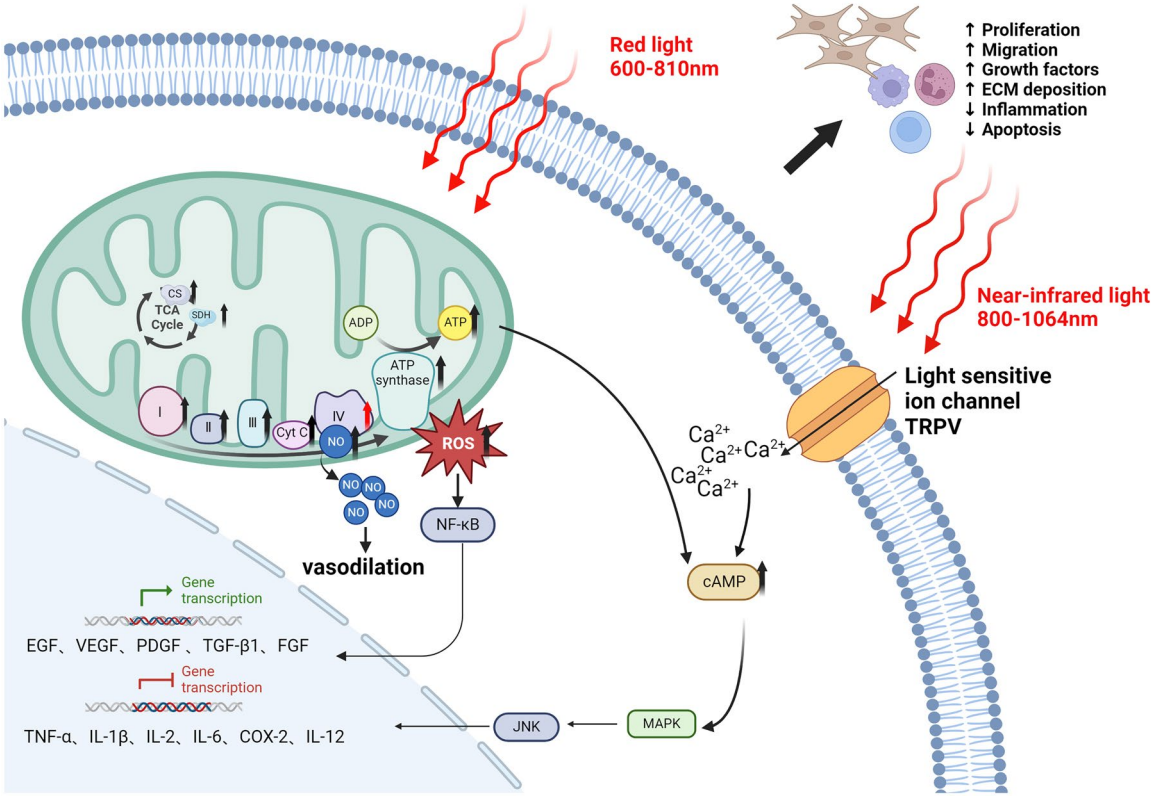


Figure 1. At the wavelengths of 600–810 nm red laser, the chromophore that absorbs photons is the CCO, which leads to upregulation of cellular respiration, a significant increase in ATP, ROS and NO. At wavelengths ranging from 800 to 1064 nm, the photosensitive channel TRPV absorbs the majority of photons, resulting in the opening of the Ca^{2+} channel on the membrane. All of these activate the second messenger. Both JNK and NF-κB were activated. Pro-inflammatory factors are reduced to inhibit inflammation and apoptosis. Increased expression of growth factors leads to fibroblast proliferation and migration. (Created in BioRender).

Various second messengers such as ROS, NO, cAMP, Ca^{2+}, etc. generated by the above pathway activate transcription factors, which then translocate into the nucleus to regulate gene transcription (it has been investigated that 68 genes showed up-regulation, while 43 genes exhibited down-regulation [19]). Meanwhile, c-Jun N-terminal kinase (JNK) phosphorylation is increased after laser irradiation, and p-JNK is able to activate the MAPK/JNK signaling pathway [20]. All of the above can reduce pro-inflammatory factors such as tumor necrosis factor-alpha (TNF-α) [21,22], interleukin-1β (IL-1β) [22], IL-2 [21], IL-6 [22], cyclooxygenase-2 (COX-2) [22], IL-12 [23] etc., which can inhibit inflammation [22] and apoptosis [21]. PBM boosts the expression of growth factors such as vascular endothelial growth factor (VEGF), epidermal growth factor (EGF), platelet-derived growth factor (PDGF), and fibroblast growth factor (FGF) [24] and transforming growth factor beta 1 (TGF-β1) [25], resulting in cell migration and proliferation of fibroblasts [26].

3. The effect and mechanism of PBM on diabetes

The danger of diabetes lies not in the diabetes itself, but the severe, and in some cases, life-threatening complications. Therefore, maintaining blood glucose within the normal range is a top priority for diabetics. Studies on PBM *in vivo* and *in vitro* for diabetes and its compilations are listed in Table 1. The widespread demonstration of the role of PBM in regulating blood glucose is now well-established in the literature. PBM can:

- Play a direct role in lowering blood glucose;
- Promote insulin secretion;
- Improve responsiveness of insulin-targeting tissues.

3.1. Lowering blood glucose

The level of blood glucose can be directly decreased, and various glucose metabolic pathways in the body can be mobilized through PBM (Photobiomodulation) irradiation of skeletal muscle or blood. Low-level laser stimulates mitochondrial components, modulates redox signaling in the respiratory chain, and affects the glycolysis, pentose phosphate pathway or gluconeogenesis pathway, as well as sucrose and starch metabolism to promote glucose metabolism.

The enhancement of mitochondrial function following PBM irradiation of skeletal muscle is also closely linked to the reduction of blood sugar levels [43–45]. Insulin-dependent glucose uptake accounts for 80% of glucose uptake by skeletal muscle [46]. In muscle tissue irradiated with PBM, giant mitochondria are formed through the fusion of small mitochondria [47]. Changes in the configuration of giant mitochondria can enhance both energy transfer and the distribution of Ca^{2+} in cytoplasm [48], which are used for skeletal muscle contractile activity and/or myoglycogen synthesis.

In addition to the formation of giant mitochondria, PBM also regulates mitochondrial enzyme activity and gene expression. PBM has the potential to regulate CCO to promote greater electron transfer within the respiratory chain and facilitate enhanced proton transport across the inner mitochondrial membrane [49,50]. PBM activates the complex I, complex II, complex III, complex IV, succinate dehydrogenase [50] and citrate synthase(CS) [51]. The PBM-regulated genes encode enzymes required in myoglycogen synthesis, such as glycogen synthase (GS) [52], glucose phosphate translocase (GPR) [53], and glucose phosphorylase (GPL) [30]. The up-regulation of the expression of these genes leads to a high level of myoglycogen synthesis. Simoes A et al. [28] found PBM can alter the carbohydrate and lipid metabolism of rats with diabetes. However, Mirian Bonifacio et al. [29] did not find a significant impact of PBM treatment on glucose homeostasis and morphometric parameters in pancreatic islets of diabetic mice. Additionally, they observed discrete impairments in glucose homeostasis due to PBM treatment.

3.2. Promotion of insulin secretion

According to a study conducted by Li et al. the levels of fasting insulin in mice exhibited a significant increase following irradiation with a compound laser on bilateral Pishu (BL 20), Shenshu (BL 23), and Sanyinjiao (SP 6) acupuncture points [30]. The laser irradiation had an impact on the pentose phosphate pathway, glycolysis, gluconeogenesis, as well as starch and sucrose metabolism. These metabolic pathways interacted with each other after the laser treatment, leading to a significant increase in the production of L-arginine. As an insulin secretagogue [54], arginine is able to activate mTOR to promote insulin synthesis [55]. Koh et al. [56] found that the insulin release induced by perfusion of high glucose combined with arginine was 2 times greater than that induced by high glucose alone or low glucose with arginine. It is suggested that in the context of diabetic hyperglycemia, the use of low-level laser irradiation to induce

Table 1. Summary of experiments *in vivo* and *in vitro*.

number	disease	Experimental tissues/cells/ animals	Light	Light parameters	mechanism	year	Ref.
1	Regulating blood sugar	Wistar rats	LED	RED (630 ± 10 nm)/25mW and IR (850 ± 20 nm)/50mW; 114 mW/cm^2 for 90 s (10 J/cm^2), 270 s (30 J/cm^2), 540 s (60 J/cm^2)	The administered PBM energy may have altered the equilibrium between the active and inactive states of the glycogen synthase enzyme, favoring the active state and enhancing glycogen synthesis.	2020	[27]
		Diabetic rats	semiconductor diode laser	660 nm, 100 mW 5, 10 and 20 J/cm^2	PBM can alter the carbohy-drate and lipid metabolism of rats with diabetes.	2009	[28]
		C57BL/6 J mice	GaAlAs diode	808 nm, 100 mW, 30 J/cm^2 (energy per point of 0.84 J)	While PBM showed positive effects on certain parameters, but there was no statistically significant difference observed. During the 4-week glucose tolerance test (GTT), mice treated with PBM exhibited discrete impairments in glucose homeostasis. Additionally, PBM treatment did not affect pancreatic morphology or adipose tissue.	2022	[29]
2	Promotes insulin secretion	type 2 diabetes mellitus (T2DM) Wistar rats	carbon dioxide laser	10.6 μm and 650 nm compound laser acupuncture	Significant reductions were observed in blood glucose and fasting insulin levels. Additionally, acupuncture and moxibustion have demonstrated effectiveness in regulating abnormal lipid metabolism.	2020	[30]
3	Insulin Resistance	HFD Male Swiss albino mice	continuous laser	infrared, 780 nm, 250mW/cm^2, 10 J/cm^2	PBM reverses decreased GLUT 4 content and AS 160 (Ser 588) phosphorylation and reduces JNK activation to promote insulin secretion.	2018	[31]
		Wistar rats	unkonwn	cover each area of 1.13 cm^2, a laser spot area of 0.028 cm^2, the total energy of 22.4 J, 70 mW, 660 nm, 20 J/cm^2, 0.56 J and 8 s per point	LPLI significantly ameliorated diabetic glycemia in rats induced by streptozotocin. Furthermore, these modifications did not coincide with alterations in insulin or HMGB1 plasma levels, instead possibly being associated with enhanced insulin sensitivity in these animals.	2016	[32]
		Wistar rats	LED	780 nm, 10 mW, 10 J/cm^2, 40 s, Total energy delivered per animal 2 J	PBM therapy mitigated the HFD-induced suppression of insulin activation.	2020	[33]
4	Diabetic Retinopathy	A total of 97 retinal explants obtained from P9 wild-type mice from 50 wild-type animals	LED	525 nm (control) and 660 nm (near infrared, NIR), 180 μW/cm^2	PBM influences retinal metabolism, enhancing mitochondrial oxidation. However, this effect alone is inadequate to ameliorate retinal structural damage induced by high glucose conditions in retinal explant cultures.	2023	[34]
		Wistar rats by STZ injection	continuous-wave diode-pumped solid-state laser	670 nm, 5mW/cm^2, 900mJ/cm^2	PBM had the ability to regulate retinal proteins and alleviate oxidative stress originating.	2021	[35]
		Transgenic mice	LED	670 nm, 40 mW/cm^2	PBM improved photoreceptor mitochondrial membrane potential, safeguarded Müller cells and photoreceptors from injury, diminishing retinal vascular leakage. The retinal laser delivery of PBM boosted mitochondrial function and offered protection against oxidative stress.	2020	[36]

(*Continued*)

Table 1. Continued.

number	disease	Experimental tissues/cells/animals	Light	Light parameters	mechanism	year	Ref.
5	Diabetic Periodontitis	C57BL/6 J wild-type mice	diode laser irradiation	810 nm, 398 mW/cm^2, 0.1 W, 0.4 mm, 4 J/cm^2, 10 s per tooth	PBM can alleviate periodontal inflammation in diabetic mice through the GLUT1/mTOR pathway, making it a potential adjunctive therapy for managing diabetic periodontitis and reducing local inflammation in hyperglycemic patients with periodontitis.	2024	[37]
6	Diabetic Foot Ulcers	Diabetic rat skin wound model	InGaAlP type diode laser	660 nm, 100 mW	PBM irradiation recruits macrophages and fibroblasts and increases fibroblast activity	2010	[38]
		Wistar rats	unkown	890 nm, 80 Hz, 3.46 J/cm^2	PBM, ha-ADS, and PBM combined with ha-ADS accelerated the proliferation phase of wound healing in rats with type 1 diabetes mellitus by modulating the inflammatory response, macrophage phenotypes, and enhancing granulation tissue formation.	2023	[39]
		Diabetic mice (C57BL/6 mice)	ALA-PDT	635 nm, 25 J/cm^2	Wounds treated with ALA-PDT showed more complete epithelial regeneration, significant reduction in microvessels, collagen volume fraction, and inflammatory cell infiltration, and significant MRSA killing.	2020	[40]
7	Diabetic Peripheral Neuropathy	neuro2A cells	low-level InGaAlP, continuous wave mode	660nm, 30mW, 1.6 J/cm^2	PBM is able to protect neurons from high glucose-induced damage by activating the AKT signaling pathway and counteracting oxidative stress	2019	[41]
		Stz-induced T1D rats	GaAs laser device	904 nm, 70 mW, 0.001 cm^2, 7000 mW/cm^2, shaved skin at a single point between L4-L5 spine levels	PBM decreased TNF-α and IL-1β levels, and p38-MAPK mRNA expression in DRG of diabetic neuropathic rats.	2022	[42]

L-arginine secretion can play a role in promoting insulin secretion and indirectly lowering glucose levels.

3.3. Improvement of insulin resistance

The glucose transporter type 4 (GLUT4) plays a crucial role in the process of insulin-stimulated glucose uptake into adipose tissue and muscle. When insulin is present, it stimulates the translocation of GLUT4 from intracellular compartments to the cell membrane, ultimately leading to an increase in glucose uptake [57]. Gabriela Silva's study [31] shows that PBM can enhance the glucose tolerance of diabetic mice on a HFD and promote the phosphorylation of Akt (Ser473). Additionally, this therapy can reverse the decreases in GLUT4 expression and the phosphorylation of AS160 (Ser588) induced by HFD. Moreover, a substantial amount of evidence suggests that inflammation caused by immune cells and adipocytes is another pathogenetic mechanism of insulin resistance in diabetes [58,59]. Immune cells and hypertrophic adipocytes secrete FFA and TNF-α. An excess of circulating FFA, TNF-α, and other factors promotes serine phosphorylation of insulin receptor substrates (IRS), causing the disruption of downstream insulin receptor (IR) signaling [60]. Amjadi et al. [61] found that PBM significantly lowered the concentrations of IL-1α, IL-1β and IL-6 in diabetic rats compared with those in non-diabetic rats.

4. The effect and mechanism of PBM on diabetes' complications

4.1. Diabetic retinopathy

Diabetic retinopathy (DR) is one of the microvascular complications of diabetes mellitus, and diabetic macular edema (DME) is the primary cause of vision loss in patients with DR. Hyperglycemia causes oxidative stress [62] and damages the retina, which is rich in easily oxidized polyunsaturated fatty acids (PUFAs) [63]. This leads to the release of various vasoactive substances (e.g. VEGF, etc.), neovascularization, and

ultimately, loss of vision. NIR radiation has been shown to enhance mitochondrial activity and ATP production in the retina which reduces free radical production and oxidative damage [64]. PBM has also been demonstrated to decrease gene expression of inflammatory mediators. Calbiague García V et al. [34] employed retinal explants to demonstrate that although photo biomodulation primarily targets mitochondrial metabolism in addressing diabetic retinopathy, they contend that this mechanism alone may not suffice to mitigate retinal structural damage induced by elevated glucose levels. Ahmed et al. [35] analyzed retinal proteins by 670 nm laser irradiation in a DR rat model and found that PBM has the potential to regulate retinal proteins like neuron specific enolase (NSE) and neuro filaments (NFs), enhance the function of retinal ganglion cells (RGCs) and photoreceptors, and decrease oxidative stress in the retina caused by DR. Shen et al. [36] discovered that LED-delivered PBM not only enhances the mitochondrial membrane potential of photoreceptors but also protects Müller cells and photoreceptors from oxidative stress damage. Additionally, PBM mitigates retinal vascular leakage [36] and reduces vascularization [65]. It opens the possibility of future clinical studies on PBM therapy in patients with retinal diseases with neovascular components, especially DME.

4.2. Diabetic periodontitis

Periodontitis highly associated with diabetes [66]. People with diabetes have a roughly threefold higher risk of periodontitis compared to those without diabetes [67], due to their intense inflammatory response to bacteria. Additionally, high blood glucose levels prevent effective wound healing, which can lead to further complications. Elevated levels of IL1-β, TNF-α, IL-6, RANKL/OPG, and oxygen metabolites have been found in the gingiva of patients with poorly controlled blood glucose, as demonstrated by animal studies [68]. As a result, diabetes triggers a more lasting inflammatory response, which will lead to attachment loss and alveolar bone resorption, and eventually imped the formation of new bone [69]. PBM has been demonstrated to be effective treatment for dysfunctional microcirculation, wound healing [70,71], pain management [72–74], fracture healing, and reduction of swelling and inflammation [75,76]. TNF-α, IL-6, and calprotectin have been identified as crucial players in the development of periodontitis and diabetes. Scaling and root planning (SRP) with PBM significantly decreased TNF-α [77], IL-6 [78] and calprotectin [79] and other pro-inflammatory proteins levels, which reduced the periodontal inflammation. Mandrillo et al. [80] examined the effect of PBM on primary human gingival fibroblast cells, with a focus on the gene activation of HAS1, ELN, DSP, ELANE, RPL13, and HYAL1, which links directly to elastin production and High-Fructose Diet (HGF) function. HGF promotes elastin deposition and collagen type I expression, which are the key proteins in matrix synthesis. Cui et al. [37] discovered that PBM can help reduce periodontal inflammation by inhibiting the GLUT1/mTOR pathway. Additionally, they found PBM impacted inflammatory macrophage senescence caused by high glucose levels by regulating glucose metabolism, thereby leading to a decrease in local inflammation in the periodontium under high-glucose conditions.

In terms of removing periodontal microplaque organisms, NIR, a kind of PBM, has been proven to effectively eliminate bacteria with no increase of tissue temperature. This suggests the potential use of PBM in decolonization without causing bacterial antibiotic resistance [81,82]. Absorption of photon energy by photosensitizers can promote free radical production to cause oxidative damage and bacterial inhibition [83]. Oxidative damage can kill deeper bacteria without damaging the tooth root [84].

Malak Fouani et al. [85] found that in streptozotocin (STZ) induced diabetes rats, the oxidative stress was increased in all three major salivary glands, regardless of the duration of the disease. Diabetic periodontitis is associated with salivary gland lipid droplet aggregation. Reduced salivary secretion impairs the 'cleansing function' of saliva. Some scholars studied the effects of PBM on the parotid gland (PG) of diabetic rats and found that red diode laser irradiation reduced lipid droplet accumulation in PGs [28,86]. In a study by Neira et al. [87], human adipocytes exposed to PBM with low-level laser energy create a transient pore in the cell membrane. This allowed the fat content to be released from inside the cells to the outside while keeping the capillaries intact.

4.3. Diabetic foot ulcers

Among people diagnosed with diabetes mellitus, the lifetime incidence of diabetic foot ulcers (DFUs) can be as high as 25% [88]. Five-year mortality following a first ulcer is about 40 per cent in people with diabetes, and ranges from 52 to 80 per cent following a major amputation [89,90]. PBM promotes tissue reconstruction of DFUs. Carvalho et al. [38] found that irradiated animals had an increased amount of collagen in the scar tissue, an increase in fibroblast proliferation and an increase in the number of capillaries formed.

Erythrocyte deformation [91] and aggregation [92] caused by high blood glucose is one of the reasons for impaired perfusion. Due to impaired perfusion in combination with peripheral neuropathy [93], foot ulcerations are often difficult to heal. In a study by Mi et al. [94], to examine erythrocyte deformability post-PBM, blood samples were prepared by introducing Ca^{2+} into normal erythrocytes from healthy individuals, thereby intentionally reducing their deformability. Laser irradiation led to diminished erythrocyte sedimentation rates and lowered blood viscosity in the samples, concurrently enhancing the deformability of the erythrocytes. Currently, there is a growing interest in studying the combined effects of stem cells and PBM in wound healing, as both therapies are highly regarded in the medical community. PBM has been proven to stimulate stem cell functions by enhancing their movement, growth, and survival. It also triggers the production of specific proteins and promotes the maturation of precursor cells [95]. Roohollah Ebrahimpour-Malekshah's research has shown that combining PBM with human allogeneic adipose-derived stem cells (ha-ADS) is more effective for wound healing than using ha-ADS or PBM alone [39]. Due to the abnormal immune function of diabetic patients [96], diabetics are particularly vulnerable to bacterial infections. And Staphylococcus aureus (MRSA) is the most commonly isolated pathogen [97] which is often poorly treated with common antibiotics. However, Huang et al. found that antimicrobial photodynamic therapy (aPDT) mediated by topical administration of 5-aminolevulinic acid (ALA) could significantly reduce MRSA burden in MRSA-infected diabetic mice, even was significantly more effective than ALA alone [40].

4.4. Diabetic peripheral neuropathy

A significant complication associated with diabetes mellitus is chronic diabetic peripheral neuropathy (DPN), which can impact approximately 50% of individuals with diabetes [98]. PBM has been demonstrated potential in stimulating axonal regeneration in injured nerves in animal studies [99]. PBM has been shown to be effective in regulating blood glucose, ameliorating pain caused by neuropathy and repairing neurons [100]. Some scholars [101] found that PBM (808 nm) down-regulated the neuronal damage marker NSE, suppressed neuroinflammation and attenuated hippocampal nerve damage in rats. The Schwann cells of peripheral nerves exhibit the presence of receptors for advanced glycation end products (RAGE), and their concentration is elevated in diabetic neuropathy [102]. It will initiate the intracellular inflammatory signaling pathway. Oxidative stress arises from an imbalance between the production of ROS and the antioxidant systems. Diabetes-induced impairments in glucose metabolism play a pivotal role in triggering oxidative stress. The imbalance causes the accumulation of toxic metabolites and excessive consumption of nicotinic acid adenine dinucleotide phosphate (NADPH), ultimately causing mitochondrial damage and excessive ROS production. This, in turn, leads to peripheral nervous system damage [103]. Using the mouse neuroblastoma cell line-N2A to construct an *in vitro* high glucose model, Victória et al. [41] discovered that PBM irradiation could suppress the expression of total AKT and elevate the cellular level of AKT phosphorylation. This modulation activated defensive and apoptosis-inhibiting processes, ultimately protecting the cells from glucose-induced injury. In diabetic rats, elevated blood sugar levels stimulate MAPK phosphorylation, particularly evidenced by the activation of p38 in peripheral sensory neurons, notably within the dorsal root ganglia [104]. PBM partially prevented the activation of p38 and reduced the sensitization of dorsal root ganglion neurons under hyperglycemic conditions, contributing to its anti-hyperalgesic effects. However, this study found that PBM had no impact on the metabolism of diabetes [42].

5. Clinical application of PBM in the treatment of diabetes and its complications

The clinical utilization of PBM in the treatment of diabetes and its associated complications has yielded promising outcomes. Research on the clinical applications of PBM for diabetes and its related disorders is detailed in Table 2.

5.1. Lower blood glucose

In 1981, Soviet scientists EN. Meschalkin and VS. Sergiewski pioneered the introduction of intravenous laser blood irradiation (ILIB) into therapeutic practices. Initially, this method was developed to treat cardiovascular diseases [131]. ILIB utilizes an intravenous catheter to puncture the patient's elbow vein and subsequently replaces the intravenous needle with a fiber-optic needle from a cannula within the intravenous catheter for endovascular laser irradiation. Kazemi et al. [105] conducted a study comparing blood samples from nine patients with type 2 diabetes before and after undergoing 632.8 nm ILBI. The results demonstrated a significant reduction in dehydroascorbic acid,

Table 2. Clinical progress of PBM in the treatment of diabetes and its complications.

number	Clinical Application	subject population	Light	Light parameters	result	year	Ref.
1	abnormally high blood glucose level	9 diabetic type 2 patients	Intravenous or intravascular laser blood irradiation	405 nm, 1.5 mW (ILBI)	There was a significant reduction in L-arginine levels in the blood of the patients. Consequently, there was a substantial decrease in blood sugar levels.	2013	[105]
		16 patients with T2 DM	LED	850 nm, 75 mW (muscle irradiation)	The combination of PBM and moderate exercise acutely lowered glucose levels among adult men with T2DM, without influencing cardiopulmonary or hemodynamic adjustments during the transition to moderate exercise.	2019	[106]
		64 obese women (BMI 30-40 kg/m2, age between 20 and 40 years old)	Ga-Al-As diode Lasers	808 nm, 100 mW (muscle irradiation)	When compared to physical training alone, physical training combined with phototherapy was better to reduce the percentage of fat mass and HOMA-IR index. Additionally, an increase in adiponectin concentration in total skeletal muscle mass was observed with the combined treatment.	2015	[107]
		10 patients with T2DM	LED	LEDs, 830 ± 20 nm (infrared) and 630 ± 10 nm (red), 25 red arrays, 80 mW each; 25 infrared arrays, 80 mW each	Following PBM treatment, the AUC significantly de-creased, indicating better glycemic control. PBM exhibited a dose- and time-response effect in reducing glycemia among patients with T2DM.	2023	[108]
2	Diabetic Retinopathy	4 eyes of 4 diabetic patients	unkown	670 nm, 25 J/cm^2, Daily PBM treatment for only 80 s per treatment twice (retinal irradiation)	PBM resulted in a significant reduction in macular thickening.	2014	[109]
		135 patients with type 1 or type 2 diabetes mellitus	unkown	670 nm, 4.5 J/cm^2, not >50 mW/cm^2 (retinal irradiation with special eye protection)	Despite being safe and well-tolerated, PBM did not demonstrate effectiveness in treating CI-DME in eyes with good vision. The findings from this trial do not justify further investigation in a phase 3 study or clinical use of PBM at this dosing frequency for DME treatment.	2021	[110]
3	Diabetic periodontitis	300 diabetic patients with chronic periodontal disease	GaAlAs laser	670 nm, 2 J/cm^2, 16 min, 5 days, 5 m W (irradiation of periodontal tissue)	When PBM was combined with conservative periodontal therapy, minimal tissue edema and in the quantity of inflammatory cells in the lamina propria was noticed. Additionally, there was a decrease in the number of blood vessels, along with distinct collagenization and homogenization of the stroma.	2013	[111]
		49 diabetic patients with chronic periodontal disease	aPDT	810 nm, 0.2 W, total energy 12 J	The combined use of aPDT + SRP is more effective in reducing periodontal pathogens than the use of a single application.	2022	[112]
		a 68-year-old diabetic male	PBM, aPDT, high-intensity	pre- and postoperative photobiomodulation (660 nm, 0.2 J/cm^2, 60 sec/site), antimicrobial photodynamic therapy of the root surface (660 nm, 0.6 J/cm^2, 30 sec, methylene blue photosensitizer), and Er:YAG root and socket debridement (2940 nm, 21 J/cm^2, 30 sec)	At the 3.5-year follow-up, the tooth remained clinically functional with radiographic resolution, signifying successful reimplantation.	2024	[113]
		60 type 2 DM patients with CP	diode laser	940nm, 20 sec Frequency, 1.061 W/cm^2	Adding a 940-nm DL to SRP did not yield additional benefits in terms of bacterial reduction compared to SRP alone for three periodontal pathogens. However, the improved clinical healing and greater reduction in HbA1c observed in the SRP + DL group may be attributed to enhanced wound healing.	2020	[114]
		22 chronic periodontitis patients with type 2 DM	GaAlAs diode laser	808 nm, 4.46 J/cm^2, 20 sec	During periodontal treatment, the application of PBM resulted in minimal additional short-term benefits for deep pocket healing in patients with type 2 DM.	2017	[115]
		21 patients with diabetic foot ulcers	pulsed near-infrared Ga-Al-As lase	808 nm, 250 mW, 8.8 J/cm^2	By restoring tissue perfusion near the wound, mitigating inflammation, and enhancing angiogenesis and vascular cell proliferation, PBM can influence surrounding cells' metabolism, ultimately facilitating wound healing.	2021	[116]

(*Continued*)

Table 2. Continued.

number	Clinical Application	subject population	Light	Light parameters	result	year	Ref.
4	Diabetic foot ulcers	16 patients with chronic leg ulcers and 16 patients with diabetic foot ulcers	PDT	570-670 nm, 50 J/cm^2 (irradiation of ulcer surface)	PBM led to a significant reduction in bacterial load, which was accompanied by an evident trend towards wound healing.	2018	[117]
		18 patients with diabetic foot	pulse laser	660 nm, 30 mW, 6 J/cm^2 (irradiation of ulcer surface)	PBM effectively shortens the tissue repair process for diabetic foot ulcers, promoting rapid healing. Additionally, in some patients who had completely lost pain and tactile sensitivity in their feet, PBM facilitated the restoration of these sensations.	2018	[118]
		80 patients with diabetic foot	GaAs laser	904 nm, 10 Jcm^{-2} OR 4 J/cm^2 OR 8 J/cm^2 (irradiation of ulcer surface)	GaAs 904 nm PBM with an energy density of 10 Jcm^{-2} /4 Jcm^{-2} /8 Jcm^{-2} can improve healing rate, area reduction, and Wagner classification of DFU.	2021	[119]
		32 patients with diabetic foot	low level laser	658 nm, 30 mW (combined with Calendula officinalis oil)	PBM, either alone or in combination with Calendula officinalis oil, effectively alleviated pain and expedited the tissue repair process in diabetic foot patients.	2016	[120]
		30 patients with grade II DFUs	Ga-As laser	904 nm, 2 J/cm^2, 90 W, given for 3 days/week for 4 weeks (11 sessions)	PBM therapy promoting angiogenesis, as evidenced by the observed decrease in serum levels of VEGF and an increase in %DWSA compared to the placebo group.	2024	[121]
		13 cases of type 2 diabetes mellitus	Intravenous laser blood irradiation	630 nm, 1.5 mW, spot size 0.01 cm^2	EGFR downregulation plays a crucial role in regulating inflammation by suppressing the EGFR/MAPK cascade after PBM.	2016	[122]
5	Diabetic peripheral neuropathy	40 Type II diabetic patients with peripheral neuropathic pain	EC laser and Thor laser	632.8 nm, 3.1 J/cm^2; 660 nm, 850 nm, 3.4 J/cm^2, 50-150 mW/cm^2, frequency 78 Hz	After PBM, there was a significant elevation in Vitamin D and Magnesium levels, accompanied by a considerable enhancement in the quality of life among patients with DPN. Additionally, notable improvements were observed in vibration perception threshold (VPT) and Michigan Neuropathy Screening Instrument (MNSI) scores, along with a reduction in Numeric Pain Rating Scale (NPRS) scores.	2019	[123]
		50 patients with diabetic peripheral neuropathy	EC laser and Thor laser	632.8 nm/660 nm/850 nm	PBM facilitated the reduction of serum NSEs in patients with diabetic peripheral neuropathy. The degree of decrease in serum NSEs was directly correlated with the improvement in the patients' quality of life.	2020	[124]
		19 patients with T2 diabetic peripheral neuropathy	EC laser, Thor Laser	EC 632.8 nm, 3.1 J/cm^2; Thor 660 nm-850 nm, 3.4 J/cm^2	PBM is effective in increasing microcirculation and reducing pain in T2DM with peripheral neuropathy.	2015	[125]
		60 DPN patients	lasers diodes	Two wavelengths of visible 630 nm and near infra-red 810 nm, peak power of 100 mW, spot diameter of 5 mm, lasted 15 min per session, 7mW, 35 mW/cm^2	60 DPN patients were divided into control and laser groups. Monofilament test results showed increased sensation in the right foot from 22 patients to 26 patients, and in the left foot from 20 patients to 25 patients. Section A scores of the Michigan questionnaire differed significantly between groups.	2023	[126]
		60 patients with DPN	novel Laser Shoe	655 nm, 830 nm IR, total power 680 mW/cm^2	Novel Laser Shoe Photobiomodulation can efficiently decrease pain of peripheral neuropathic. And average and maximum plantar pressure distribution was significantly reduced.	2024	[127]
		40 diabetic patients with TN	He-Ne laser	15 mill watt helium-neon, 830 nm, 50-170 mw/cm^2, lasted 20 min	PBM and EMT, as assessed in this study, can effectively treat TN in diabetes patients. However, laser therapy demonstrates greater efficacy in alleviating pain and enhancing CMAP.	2023	[128]
		30 participants with type 2 diabetes and painful DPN	LED	890nm, 30 min, 58.5 J/cm^2/min, 3 times a week, for 12 weeks	PBM therapy does not increase IENFD over short-term usage. However, MIRE therapy provides symptomatic benefit and improves QoL in patients with painful DPN.	2021	[129]
		60 individuals with diabetes	Continuous LED	890nm, 1.3 J · cm^{-2} · min^{-1}	PBM therapy did not yield greater enhancements in peripheral sensation, balance, pain relief, or overall quality of life compared to sham therapy.	2008	[130]

L-histidine, R-3-hydroxybutyric acid, glucose-6-phosphate and L-alanine levels. Additionally, there was a notable increase in L-arginine levels. Notably, the blood glucose levels of the patients also decreased significantly. Due to the invasive nature of ILIB, laser irradiation of skeletal muscle is more commonly used in clinical practice. Oliveira et al. [106] employed laser irradiation on skeletal muscles in patients and discovered that when combined with moderate-intensity exercise (15 min, including 3 min of warm-up, 6 min of moderate intensity, and 6 min of cooldown), PBM resulted in a significant reduction in blood glucose levels. Notably, the PBM did not impact lactate levels or hemodynamics. A controlled trial revealed that the combination of physical training and phototherapy was more effective than physical training alone in reducing obesity, increasing muscle mass, and improving insulin resistance [107]. In the absence of combined PBM, the glucose-lowering effect did not appear until after prolonged aerobic exercise (30–70 min) [132]. Scontri et al. [108] explored, for the first time, the application of PBM alone or in combination with oral hypoglycemic medication in patients with T2DM. Their findings demonstrated that PBM alone or in conjunction with oral hypoglycemic medication outperformed oral hypoglycemic medication alone in reducing blood glucose levels, notably resulting in a lower area under the curve (AUC) of the blood glucose curve. These results suggest promising avenues for future clinical research.

5.2. Diabetic retinopathy

To manage DME effectively, apart from stringent glycemic control, a multidisciplinary approach is necessary, encompassing intravitreal anti-VEGF agents, corticosteroids, laser photocoagulation, and vitrectomy [133]. As research in this area continues to break through, anti-VEGF drug injections have greatly improved treatment outcomes and have become the first line of therapy [134]. While anti-VEGF therapy shows promise, it still does not address the underlying issue of macular hypoxia [135]. Additional intravitreal injections are often necessary as the concentration of VEGF recovers. However, multiple intravitreal injections can lead to numerous side effects [136]. PBM has garnered significant attention from researchers due to its non-invasive nature and low-level characteristics, with many ongoing clinical trials. Johnny Tang et al. [109] employed PBM to treat four eyes of four patients, while the contralateral eye served as an untreated control. The outcomes revealed a noteworthy decrease of focal retinal thickening with no adverse effects linked to the treatment throughout follow-up. However, in a phase II randomized clinical trial, although PBM given to participants by the investigators was safe and well tolerated, no significant therapeutic effect for the treatment of Central Involved Diabetic Macular Edema (CI-DME) in eyes with good vision [110].

5.3. Diabetic periodontitis

Mechanical methods SRP supplemented with antibiotics are currently the conventional method for treating chronic periodontitis (CP). Mechanical elimination can significantly decrease the microbial load in the subgingival area. However, due to the intricate anatomy of the root, it may not completely eradicate all pathogens [137]. Long-term subgingival planning may easily lead to root damage, while prolonged antibiotic use can increase the risk of toxicity, resistance, and other adverse effects [138]. Additionally, diabetic patients may require different/additional periodontitis treatments due to the subgingival microbiome differences between T2DM patients and systemically healthy subjects [139].

aPDT is currently used as an excellent antimicrobial therapy, with various lasers including neodymium: yttrium aluminum chal-cocobite (Nd:YAG) lasers, neodymium doped yttrium aluminum garnet (Nd:YAG) lasers, erbium and chromium doped yttrium scandium gallium garnet (Er,Cr:YSGG) lasers, erbium: yttrium aluminum garnet (Er:YAG) lasers, carbon dioxide (CO_2) lasers, and diode lasers [140]. Susyane Vieira Oliveira [141] suggests that PBM stimulation of the salivary glands has benefits in the prevention of hyposalivation and changes in salivary flow resulting from lesions by radio-therapy, especially for outcomes of stimulated salivary flow, pain, and prevention of the severity of hyposalivation. According to Susyane Vieira Oliveira, PBM stimulation of the salivary glands has been found to be beneficial in preventing hyposalivation and alterations in salivary flow caused by radiotherapy-induced lesions. This treatment has shown positive outcomes in terms of stimulated salivary flow, pain relief, and reducing the severity of hyposalivation, indirectly supporting the use of PBM in treating diabetic periodontitis. Obradović R et al. [111] found that PBM combined with conservative periodontal therapy resulted in the disappearance of tissue edema, minimal inflammatory cells in the lamina propria, reduced blood vessels, and improved collagenization and homogenization of the stroma. Multiple clinical trials have demonstrated that PBM in combination with scaling and SRP is more

effective in reducing inflammation and controlling periodontal bacterial growth. Sufaru et al. [112] discovered that both SRP alone and SRP combined with aPDT significantly reduced plaque index (PI), bleeding on probing index (BOP), probing depth (PD), and clinical attachment loss (CAL) in patients with T2DM and periodontitis, compared to SRP alone. Intentional replantation (IR) is an emerging and cost-effective last-resort treatment for persistent apical periodontitis. Alex Simon [113] reported a novel phototherapy-assisted IR protocol conducted on a compromised lateral incisor with an extensive periapical infection (Ø > 10 mm) in a 68-year-old diabetic male. The tooth at 3.5-year follow-up remained clinically functional with radiographic resolution of the infection indicating a successful reimplantation. There were Meta-analysis [142] evaluating HbA1c% and fasting glucose (FPG) levels of patients after treatment with several therapies, including SRP, SRP + antibiotic, SRP + aPDT + doxycycline (Doxy) and SRP + laser. The findings suggest that periodontal therapy combining aPDT with Doxy may be the most effective approach in reducing HbA1c% among non-smoking patients with CP. It implied that phototherapy combined with pharmacologic and mechanical means to assist in the treatment of CP is more effective, accompanied by the effect of lowering blood glucose. However, Kocak, E. et al. [114] found that the addition of a 940-nm DL to SRP did not yield additional benefits in terms of bacterial reduction compared to SRP alone for three periodontal pathogens. Oya Demirturk-Gocgun et al. [115] observed that the utilization of PBM during periodontal treatment yielded minimal additional short-term benefits for deep pocket healing in individuals with T2DM.

5.4. Diabetic foot ulcers

Commonly utilized treatments for DFU encompass glycemic control, wound debridement, vascular surgery, antibiotic therapy (both topical and systemic), pressure offloading, and wound dressings that hydrogel, encompass silver, alginate, foam dressings and hydrocolloids [143]. Despite these interventions, approximately 40%–70% of chronic ulcer patients remain unhealed and eventually face amputation [144]. PBM is a highly promising treatment for DFU, with the unique advantages of being highly selective, having fewer side effects, being non-resistant to bacteria, and being reproducible. No adverse events related to the treatment were reported [145]. In most studies, regardless of the characteristics of the selected laser (intensity, wavelength, irradiated area, etc.), PBM has been shown to improve DFU cure rates compared with standard wound care alone. A prospective clinical controlled study [116] has used daily photo biomodulation therapy (pulsed near-infrared 808 nm Ga-Al-As laser, 250 mW, 8.8 J/cm^2) in patients with debilitating DFUs at their own home. The study revealed a substantial decrease in ulcer dimensions, with no reported adverse effects associated with the device usage. The application of PBM (660 nm, 6 J/cm^2) to DFU has been shown to promote rapid proliferation of granulation tissue and skin epithelialization, resulting in viable edges that are free of maceration, keratosis, and devitalized tissue, thereby facilitating tissue repair [118]. Cardoso et al. [119] were the first to apply three different energy density laser levels to study DFU wound healing, and they found that GaAs PBM all significantly favored healing rate, ulcer area reduction, and Wagner classification of DFU.

Recent clinical research indicates that one of the mechanisms by which GaAs promotes DFU is by stimulating angiogenesis [121]. Carvalho et al. [120] applied 658 nm PBM in combination with Calendula officinalis for the treatment of diabetic foot ulcers and found that the combined use of PBM was effective in relieving pain associated with diabetic foot due to its anti-inflammatory action and its ability to accelerate the tissue repair process.

5.5. Diabetic peripheral neuropathy

None of the numerous drugs that have targeted the underlying molecular and cellular mechanisms of DPN have been approved by the U.S. Food and Drug Administration (FDA) for the treatment, prevention, and mitigation of DPN [146]. Glycemic control remains the only effective tool for managing DPN. PBM has been clinically proven to directly reduce neuronal inflammation and alleviate patient suffering. N Kazemikhoo et al. [122] found that by analyzing blood samples from diabetic patients before and after intravenous laser therapy, it may reduce neuroinflammation and secondary damage by inhibiting the EGFR/MAPK cascade. Anju et al. [123] conducted an experiment on 40 patients with DPN and found after undergoing PBM, there was a notable elevation in the levels of vitamin D and magnesium. without any magnesium supplementation. PBM-induced increases in vitamin D and magnesium levels could promote neuronal repair [123]. Additionally, they observed a decrease in pain values among the DPN patients. They also studied how PBM affects the levels of NSE in the blood of patients with DPN. The findings revealed enhanced

quality of life along with a reduction in serum NSE levels [124]. It has been also found a significant increase in temperature at the site of laser treatment in DPN patients, concluding that microcirculation was improved at the site of PBM-irradiated neurons [125]. Neuropathy is often associated with heightened plantar pressure and prolonged stance duration [147]. Ebadi SA et al. [126] administered PBM with wavelengths of 630 nm and 810 nm to irradiate surface of each patients' foot. Results demonstrated improved sensation in both feet, as indicated by the monofilament test and Michigan Neuropathy Screening Instrument (MNSI) scores. G Arun Maiya et al. [127] were the first to utilize Laser Shoe on DPN patients, observing improvements in plantar pressure and pain relief in T2DM individuals. PBM can reorganize disrupted muscle force patterns caused by musculoskeletal dysfunction, thus enhancing gait patterns in DPN patients. A randomized controlled trial found both electromagnetic therapy (EMT) and PBM effective in treating trigeminal neuralgia (TN) in diabetic patients. PBM showed superior efficacy in pain relief and compound muscle action potential (CMAP) improvement [128]. However, some studies have found that the efficacy of PBM therapy is not significantly different from that of a placebo [129,130].

6. Discussion

PBM has been shown to lower blood glucose. It is currently recognized as ILBI and skeletal muscle laser irradiation. Since ILBI is an invasive treatment, moderate-intensity exercise combined with skeletal muscle laser irradiation may be a more favorable long-term treatment option. But there are studies suggest that non-vascular irradiation may not necessarily lead to significant improvements in blood glucose metabolism [29, 42]. The majority of individuals with DFUs often need to make frequent visits to hospital clinics lasting several weeks, presenting significant logistical challenges and safety issues, particularly during periods of self-isolation and social distancing, especially for frail elderly patients. Meanwhile, experts support the idea that providing PBM treatment at home in conjunction with standard care is a viable and risk-free alternative for individuals with DFUs and co-morbidities [116]. In the future, PBM holds potential to emerge as a portable, minimally invasive, user-friendly, and cost-efficient therapeutic tool for DFU [148]. However, PBM has not yet been widely used in clinical practice, whether it has stable and predictable therapeutic effects still needs to be further explored.

There have been numerous studies that focus on the use of PBM in combination with conventional therapies, all of which have shown better efficacy of the combination therapy in the treatment of diabetes mellitus and its complications. Among the PBM treatments for diabetes and its complications, there is a relatively high number of reports on the treatment of diabetic foot ulcers and diabetic periodontitis, both belongs to tissue regeneration. Currently, there are many studies combining the use of stem cells and PBM. In the future, regenerative stem cell laser therapy may play a crucial role in rejuvenating or replacing damaged body tissues and organs in a customized manner. There has been very little literature on PBM administered alone for the treatment of periodontitis, both domestically and internationally. Most studies focus on PBM used in conjunction with drugs or SRP. The misuse of antibiotics has led to a significant problem with drug-resistant bacteria in the clinical treatment of periodontitis. PBM, due to its non-toxicity, efficacy, lack of resistance, and specificity to diabetic patients, can be used in conjunction with the traditional therapy of SRP to lower the patient's blood glucose level while safely and effectively sterilizing the bacteria. Antimicrobial photodynamic therapy with topical 5-aminolevulinic acid (ALA-PDT) has been shown to be highly effective in killing MRSA [40]. 658 nm PBM combined with Calendula officinalis for diabetic foot ulcers demonstrated the safety and efficacy of PBM in combination with traditional Chinese herbs [120]. The low power and low energy of PBM, when used alone, will inevitably result in a weak therapeutic effect, so it is mostly used as an adjunctive therapy in clinical practice. However, there are also studies that have not found benefits associated with combined PBM therapy [114,115].

PBM is a non-invasive treatment modality that has demonstrated no adverse effects in the treatment of diabetic complications. In contrast, laser photocoagulation, a laser-based eye surgery used to treat DME, aims to destroy abnormal blood vessels and prevent further vision loss. Although effective in treating several eye diseases, it acts directly on the retina and may lead to adverse events such as vision loss and decreased color vision [149]. And some clinical trials have not observed a benefit of PBM for DME [110]. To address these limitations, subthreshold micro pulse laser therapy (SML) has been developed as a form of PBM. It employs a longer cooling period to prevent retinal burns while achieving better visual acuity compared to conventional laser therapy [150]. Furthermore, studies have shown that combining SML with anti-VEGF injections can enhance the efficacy of anti-VEGF drugs, reducing

the number of intravitreal injections needed [151–153]. This approach not only lowers treatment costs but also minimizes the number of follow-up visits and the adverse effects associated with fundus injections. Thus, SML holds promising potential as a complementary therapy in the management of diabetic eye diseases.

While PBM holds promise as a treatment for diabetes and its complications, some studies have not observed significant therapeutic effects [29, 34, 42, 110, 114,115, 129,130]. The limited efficacy of PBM therapy may be attributed to various factors. Firstly, as noted in many randomized clinical trials, the therapeutic parameters utilized in the study may require further optimization and unification, including frequency, dosage, or wavelength. A unique aspect of PBM therapy is the dose response. Most biological interventions follow the usual dose-escalation path with an optimum threshold dose, beyond which there is a plateau or toxicity. Secondly, existing studies' inclusion criteria (animals or patients) are uneven, which could impact the assessment of PBM therapy effectiveness (Light absorption capacity and mitochondrial status are different in each individual). PBM parameters that work in animals may not work in humans. Additionally, not every controlled study has scientifically used placebos. And a critical technical hurdle in the realm of PBM research is the scarcity of appropriate devices to enhance the optimization of studies. Recently, in addition to specialized light applicators and LED arrays, other devices have emerged for PBM irradiation, such as laser eye protection [110]. Considering these factors, further evaluation of PBM therapy efficacy may necessitate more detailed and comprehensive studies, involving optimization of treatment parameters and devices, broadening the sample range, and considering other potential influencing factors.

7. Conclusion

Diabetes mellitus is one of the most serious chronic illnesses in the world. Clinically, PBM has shown significant benefits in regulating circadian rhythms and managing skin disorders like wrinkles, acne scars, hypertrophic scars, and burn healing. Its applications extend to diabetes mellitus and its related complications. PBM is either replacing or being used in conjunction with traditional therapies, owing to its non-invasive nature and minimal side effects. While the mechanism of PBM is being actively researched, its clinical utility and value are gaining widespread recognition. However, despite the wealth of research on laser therapy, a unified standard for its application remains elusive. Unknown parameters such as laser wavelength, energy, irradiation time, and auxiliary drugs are still in the exploratory stage, lacking data and theoretical support. Future studies could aim at optimizing laser therapy to provide the most suitable parameters of low-level laser for different diseases.

Author contributions statement

Keyan Wang and Hongwei Zhao were involved in the conception and design of the paper, drafting it, and critically revising it for intellectual content; Jingyan Ge and Yan Cheng were involved in the final approval of the version to be published; Wei Zhang, Xiaoqing Zhao, and Xiaoyu Zhang were involved in the analysis and interpretation of the data; and all authors agree to be accountable for all aspects of the work.

Disclosure statement

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Data availability statement

Data sharing not applicable - no new data generated.

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